



NASTP INSTITUTE OF INFORMATION TECHNOLOGY
Department of Computer Science | BS Computer Science

AI335L – Deep Learning Lab

Sketch-to-Face Image Synthesis

Using Pix2Pix Conditional Generative Adversarial Network

"Turning forensic sketches into photorealistic faces through end-to-end deep learning."

Course Code	AI335L – Deep Learning Lab
Instructor	Lecturer Haseeb
Semester	V (Spring 2024)
Submission Date	April 27, 2026
GitHub Repository	github.com/BILAL-ASIF-github/Sketch-to-Face-Image-Synthesis

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Section 2: Executive Summary / Abstract

Forensic face reconstruction—converting a rough witness sketch into a photorealistic portrait—has long been a manual, time-intensive process subject to artistic bias. This project, Sketch-to-Face Image Synthesis using Pix2Pix cGAN, automates that process using an end-to-end deep learning pipeline trained on paired edge-map and face-image data.

The core of the system is a Pix2Pix Conditional GAN built around a U-Net encoder-decoder generator and a PatchGAN discriminator. Beyond this generative backbone, the system integrates a full spectrum of deep learning techniques required by the course: a Feedforward MLP serves as the baseline model; a CNN-based feature extractor pre-processes and enhances edge maps; an LSTM module captures sequential structural priors from multi-step sketch refinement; and a Transformer-based perceptual loss head computes semantic similarity between generated and real faces. Transfer learning is applied via a pretrained VGG-16 backbone frozen up to layer conv4_3 and fine-tuned thereafter for perceptual loss computation.

Training data is sourced from the CelebA dataset (202,599 face images) paired with Canny-generated edge maps, augmented with the CUHK Face Sketch Database for hand-drawn sketch realism. Preprocessing includes Gaussian blurring, Canny edge extraction, normalization, and random flips/rotations. Regularization combines Dropout (0.5 in decoder layers), BatchNorm, and Early Stopping. Optimizer comparisons (Adam vs. RMSProp) and Optuna-based hyperparameter tuning are built into the training pipeline. The final system is deployed as a Streamlit web application where users upload a sketch and receive a generated face in real time.

Expected outcomes include a generator achieving $FID \leq 50$ on a held-out test set, a working deployed demo, and a reproducible codebase with pinned dependencies. This project demonstrates mastery of generative modelling, transfer learning, regularization, and system deployment within a single coherent application.

Section 3: Problem Statement & Motivation

Problem Overview

In many criminal investigations, there is no photographic or video evidence of a suspect. In such cases, law enforcement agencies rely on forensic sketches created from eyewitness descriptions. These sketches are then used to identify suspects manually or through limited software tools.

However, this process suffers from three major issues:

- Subjectivity of human memory and artistic interpretation
- Low recognition accuracy when matching sketches to real faces
- Slow investigation timelines, especially with large criminal databases

According to reports by organizations like the National Institute of Justice, forensic sketches are used in a significant portion of violent crime investigations, yet their effectiveness remains limited. Studies show that manual identification accuracy from sketches ranges between 30% to 40% under controlled conditions, and often drops further in real-world scenarios.

This creates a critical gap between sketch-based evidence and reliable suspect identification.

Who Is Affected

The limitations of sketch-based identification impact multiple stakeholders:

- | | |
|---|---------------|
| • Law Enforcement | Agencies |
| Struggle to identify suspects quickly when only sketches are available. | |
| • Missing Persons | Organizations |
| Face difficulty reconstructing faces from limited visual information. | |
| • Forensic Artists & Investigators | Investigators |
| Depend on manual processes that are time-consuming and inconsistent. | |
| • Public Safety | Systems |
| Delays in identification allow suspects to evade capture longer, increasing risk. | |

In large-scale systems like the FBI Next Generation Identification, which contains tens of millions of records, searching for a suspect without a proper photographic reference becomes extremely inefficient.

Existing Solutions

A. Traditional Methods

- Hand-drawn forensic sketches by artists
- Composite software like FACES, E-FIT

Limitations:

- Low realism
- Highly dependent on artist skill
- Poor recognition rates

B. Deep Learning-Based Generation

- Models like Pix2Pix and CycleGAN convert sketches to images
- Diffusion models generate faces from prompts

Limitations:

- Lack of structural accuracy in faces
- Poor handling of fine details like hair and facial features
- Not optimized for forensic-level precision

Face Recognition Systems

- Tools like InsightFace
- Embedding-based matching using cosine similarity

Limitations:

- Designed for photo-to-photo matching, not sketch-to-photo
 - Performance drops significantly when input is a sketch
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Integrated Systems (Recent Work)

Some recent systems combine:

- Pix2Pix for generation
- Face embeddings
- FAISS-based search
- CCTV integration

Limitations:

- Use small or synthetic datasets (e.g., ~4,000 images)
 - Rely on basic GAN architectures only
 - Lack advanced feature learning (no attention, no sequence modeling)
 - Limited generalization to real-world sketches
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4. Key Limitations in Current Approaches

Across all existing methods, the main shortcomings are:

- Low realism in generated faces
- Weak identity preservation
- Poor generalization across sketch styles
- Limited use of advanced deep learning architectures
- Small or non-diverse datasets
- Lack of robust preprocessing for real sketches

Most systems either:

- Focus on generation only, OR
- Focus on recognition only

Very few systems address both quality generation and structural understanding of sketches.

5. Motivation for This Project

The motivation behind this project is to bridge the gap between sketch-based inputs and photorealistic facial reconstruction using advanced deep learning techniques.

Instead of relying solely on basic GAN models, this project integrates multiple architectures to improve both:

- Structural understanding of sketches
- Visual realism of generated faces

By leveraging:

- Large-scale datasets (CelebA, CUFS)
- Multi-stage processing (CNN, LSTM, GAN, Transformer)
- Transfer learning with VGG-16

the system aims to produce more accurate, detailed, and consistent facial reconstructions compared to existing approaches.

Gap Statement

Current forensic sketch-to-face systems either lack realism due to simplistic generative models or fail to generalize due to limited datasets and weak architectural design. Additionally, existing approaches do not effectively capture both the structural and semantic features required for accurate face reconstruction. This project addresses these gaps by developing a multi-architecture deep learning pipeline that combines convolutional, sequential, generative, and attention-based models to significantly enhance the quality and reliability of sketch-to-face synthesis, making it more suitable for real-world forensic applications.

Section 4: Literature Review

4.1 Foundational Papers

This section reviews existing research in sketch-to-face synthesis, facial recognition, and AI-based forensic systems. The selected works include foundational architectures, recent advancements, and supporting technical contributions. Each study is analyzed based on its problem, methodology, dataset, results, and limitations.

4.1 Foundational Papers

1. Isola et al. (2017) – Pix2Pix (cGAN for Image-to-Image Translation)

- **Problem** **Addressed:**
Traditional image translation tasks lacked a generalized framework for mapping one visual domain to another, such as sketches to photos.
 - **Approach:**
Introduced a Conditional GAN (cGAN) framework using:
 - U-Net generator with skip connections
 - PatchGAN discriminator for local realism
 - Combined adversarial + L1 loss
 - **Dataset** **Used:**
Multiple paired datasets including edges-to-shoes, maps-to-aerial images.
 - **Results:**
Achieved high-quality photorealistic outputs with strong structural preservation in paired image translation tasks.
 - **Limitation:**
Requires paired datasets, which are difficult to obtain in real forensic scenarios. Also struggles with generalization across diverse sketch styles.
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2. Yang et al. (2022) – S2FGAN (Semantically Aware Sketch-to-Face GAN)

- **Problem** **Addressed:**
Lack of semantic control in sketch-to-face generation, leading to unrealistic outputs.
 - **Approach:**
Proposed S2FGAN with:
 - Semantic segmentation maps
 - Interactive control over facial attributes
 - GAN-based architecture with attention mechanisms
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- **Dataset** **Used:**
CUFS and CelebA datasets.
 - **Results:**
Improved realism and controllability of generated faces, especially for attributes like hair, eyes, and accessories.
 - **Limitation:**
Still dependent on structured inputs and lacks robustness in handling free-hand forensic sketches.
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4.2 Recent Papers (Last 3 Years)

3. ACM (2025) – AI-Based Sketch-to-Real-Time Criminal Identification System

- **Problem** **Addressed:**
Manual suspect identification from sketches is slow and inaccurate, with ~30–40% success rate.
 - **Approach:**
End-to-end pipeline:
 - Pix2Pix for sketch-to-photo synthesis
 - InsightFace for feature embedding
 - FAISS for similarity search
 - Real-time CCTV integration
 - **Dataset** **Used:**
~4,500 synthetic face images (Kaggle), converted into sketches.
 - **Results:**
 - MSE: 0.0025
 - SSIM: 0.8989
 - FID: 16.85
 - Identity similarity: 0.7391
Improved investigation speed and matching accuracy.
 - **Limitation:**
Uses synthetic dataset, limited real-world diversity, and lacks large-scale deployment validation.
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4. Pathak et al. (2025) – DSFace (Diffusion-Based Sketch-to-Face)

- **Problem** **Addressed:**
GANs struggle with stability and fine details, while diffusion models lack structural control.
 - **Approach:**
Two-stage hybrid system:
 - Stage 1: GAN generates coarse face
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- Stage 2: Latent diffusion with ControlNet + DINO-V2 embeddings
 - Reformulates problem as conditional inpainting
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- **Dataset** **Used:**
CUFS dataset (606 identities).
 - **Results:**
 - SR-SIM: 0.8941
 - MSSIM: 0.8618
 - FID: 62.58 (best among compared models)
 - IS: 1.49
Achieved superior realism and structural consistency.
 - **Limitation:**
High computational cost, complex architecture, and slower inference, limiting real-time deployment.
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4.3 Additional Supporting Sources

5. Yoshikawa et al. (2022) – Sketch-to-Diverse Image Generation

- **Problem** **Addressed:**
Single-output generation limits diversity in sketch-to-image synthesis.
 - **Approach:**
Proposed probabilistic generation framework:
 - Generates multiple outputs from one sketch
 - Uses latent space sampling
 - **Dataset** **Used:**
Sketchy and CUFS datasets.
 - **Results:**
Produced diverse and realistic variations of images from a single sketch.
 - **Limitation:**
Diversity comes at the cost of identity preservation, which is critical in forensic applications.
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6. IJIRCST (2021) – Face Sketch Recognition System

- **Problem** **Addressed:**
Difficulty in matching sketches with photos using traditional ML methods.
 - **Approach:**
Used:
 - Feature extraction techniques (HOG, SIFT)
 - Classical classifiers (SVM)
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- **Dataset** **Used:**
Small-scale face sketch datasets.
- **Results:**
Achieved moderate accuracy in controlled environments.
- **Limitation:**
Poor scalability, low accuracy on real-world sketches, and inability to handle large databases.

7. ArXiv (2024) – Deep Generative Models for Sketch-to-Face

- **Problem** **Addressed:**
Need for improved realism and generalization in sketch-to-image translation.
- **Approach:**
Compared GANs vs diffusion models:
 - Introduced hybrid generative pipelines
 - Focused on improving perceptual quality
- **Dataset** **Used:**
CelebA, CUFS, and synthetic datasets.
- **Results:**
Diffusion-based models outperformed GANs in realism and perceptual metrics.
- **Limitation:**
Computationally expensive and not suitable for real-time forensic systems.

Study Model	Approach	Dataset	Strengths	Limitations
Pix2Pix (2017)	cGAN (paired translation)	Paired datasets	Fast, strong baseline	Needs paired data, weak identity preservation
S2FGAN (2022)	Semantic GAN	CUFS, CelebA	Better facial structure	Requires annotations, complex
ACM System (2025)	GAN InsightFace FAISS	+ Synthetic (4,500 images)	End-to-end pipeline	Limited real-world generalization
DSFace (2025)	Diffusion ControlNet	+ CUFS	High realism, best metrics	High cost, not real-time
Yoshikawa et al. (2022)	Multi-output GAN	Sketch datasets	Output diversity	No identity guarantee
IJIRCST (2023)	CNN recognition	Standard datasets	Good classification	No generation capability
Diffusion Models (2024)	Denoising models	Large datasets	Best visual quality	Slow and expensive
Proposed Model (Your Work)	Pix2Pix cGAN + InsightFace	+ Custom synthetic processed	Complete + real-time scalable	pipeline, capable, search, Depends on dataset quality, limited real-

Study Model	Approach	Dataset	Strengths	Limitations
	FAISS + Real-time CCTV	dataset images)	(4,500 balanced and speed	accuracy sketch training, GAN realism constraints

Gap Identification (Strong Closing for Defense)

Existing research either focuses on **high-quality image generation** or **accurate face recognition**, but rarely integrates both into a **single practical system**. GAN-based approaches are efficient but lack realism, while diffusion-based methods provide superior quality but are computationally expensive and unsuitable for real-time forensic deployment.

This project addresses this gap by proposing a unified, end-to-end pipeline that combines:

- Efficient sketch-to-image generation (Pix2Pix)
- Robust identity extraction (InsightFace)
- Scalable similarity search (FAISS)
- Real-time surveillance integration

This balance between **accuracy, speed, and deployability** makes the proposed system more practical for real-world forensic applications compared to existing approaches.

Section 5: Project Objectives & Scope

5.1 SMART Objectives

5.1 Project Objectives (SMART Framework)

Objective 1: Develop Sketch-to-Face Generation Module

Design and implement a Pix2Pix-based conditional GAN model that converts forensic sketches into realistic facial images.

- **Specific:** Convert grayscale sketches into color facial images
 - **Measurable:** Achieve $SSIM \geq 0.85$ and $FID \leq 20$ on test dataset
 - **Achievable:** Using existing architectures like Pix2Pix and prepared paired dataset
 - **Relevant:** Core requirement for enabling downstream identification
 - **Time-bound:** Completed within 3 weeks (Weeks 2–4)
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Objective 2: Implement Facial Feature Extraction & Matching System

Develop a system using InsightFace embeddings and FAISS similarity search to match generated faces with stored criminal records.

- **Specific:** Extract embeddings and retrieve top-k similar faces
 - **Measurable:** Achieve $\geq 70\%$ matching accuracy (Top-5 retrieval)
 - **Achievable:** Using pre-trained InsightFace and FAISS indexing
 - **Relevant:** Enables automated suspect identification
 - **Time-bound:** Completed within 2 weeks (Weeks 5–6)
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Objective 3: Integrate Real-Time Detection Module

Build a real-time CCTV-based detection system using OpenCV to identify suspects from live video streams.

- **Specific:** Detect and match faces in live video feed
 - **Measurable:** Achieve real-time performance ≥ 15 FPS with detection accuracy $\geq 65\%$
 - **Achievable:** Using optimized OpenCV + lightweight inference pipeline
 - **Relevant:** Adds practical real-world application
 - **Time-bound:** Completed within 2 weeks (Weeks 7–8)
-

Objective 4: Develop End-to-End Integrated Pipeline

Combine sketch generation, recognition, and detection into a single working system.

- **Specific:** Build a unified workflow from sketch input to suspect identification
 - **Measurable:** Reduce identification time by $\geq 40\%$ compared to manual process (simulated)
 - **Achievable:** By integrating previously developed modules
 - **Relevant:** Addresses real forensic workflow inefficiencies
 - **Time-bound:** Completed within Week 9
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Objective 5: Evaluate and Validate System Performance

Conduct quantitative evaluation using standard metrics and compare with baseline models.

- **Specific:** Evaluate using MSE, SSIM, FID, and cosine similarity
 - **Measurable:** Demonstrate improvement over baseline (CycleGAN or basic CNN methods)
 - **Achievable:** Using test dataset and existing evaluation metrics
 - **Relevant:** Required for academic validation
 - **Time-bound:** Completed within Week 10
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5.2 Project Scope**In Scope**

The project will include:

- Sketch-to-face image generation using Pix2Pix GAN
 - Facial feature extraction using deep learning embeddings (InsightFace)
 - Similarity search using FAISS indexing
 - Real-time face detection using OpenCV
 - Integration into a complete pipeline system
 - Evaluation using standard image quality and recognition metrics
 - Use of synthetic and preprocessed datasets
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Out of Scope

To keep the project feasible and focused, the following are explicitly excluded:

- Use of real criminal databases or sensitive law enforcement data
 - Training on large-scale real-world sketch datasets due to availability constraints
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- **Implementation of advanced diffusion models (due to high computational cost)**
 - **Deployment on cloud or large-scale production environments**
 - **Integration with actual government or surveillance infrastructure**
 - **Handling legal, ethical, or privacy compliance frameworks in depth**
 - **Mobile application or large-scale UI/UX product development**
 - **Multi-modal inputs (e.g., text-to-face or voice-to-face generation)**
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5.3 Scope Justification (Defense Point)

The scope is intentionally constrained to ensure:

- **Technical depth over unnecessary breadth**
 - **Completion within academic timeline**
 - **Focus on a working prototype rather than theoretical exploration**
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By clearly defining boundaries, the project avoids common risks such as scope creep, incomplete modules, and lack of evaluation, ensuring a fully functional and defensible system by the final submission.

Section 6: Dataset Description

Primary Dataset: CelebA

Data Source & Licensing

The primary dataset used in this project is the **CelebA Dataset**, developed by **Chinese University of Hong Kong**. It is publicly available for **research and academic purposes**, making it legally usable for this project.

Size, Format, and Modality

- **Total Images:** ~202,599 face images
 - **Identities:** 10,177 individuals
 - **Image Format:** JPEG
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- **Resolution:** $\sim 178 \times 218$ pixels
 - **Modality:** RGB facial images
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For this project:

- Sketches will be **synthetically generated** from images
 - Final modality becomes:
 - Input: Sketch images
 - Output: RGB face images
-

Class Distribution

Since this is not a traditional classification problem, class distribution is handled differently:

- **Task Type:** Image-to-image translation (Sketch $\rightarrow$ Face)
 - Each image pair represents one training sample
 - Distribution is:
 - Balanced in terms of identity diversity
 - But **biased toward celebrities (controlled lighting, poses)**
-

👉 If converted into classification (identity matching), the dataset is **highly imbalanced per identity**.

Quality Issues & Challenges

Several real-world issues exist:

- **Domain Gap**
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- Synthetic sketches vs real forensic sketches
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- **Bias**

- Overrepresentation of celebrities, limited demographic diversity
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- **Noise**

- Variations in pose, lighting, occlusion (glasses, hair)
-

- **Lack of Real Sketches**

- Requires artificial sketch generation
-

- **Overfitting Risk**

- Clean dataset may not generalize to real-world cases
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Preprocessing Pipeline

A structured preprocessing pipeline will be applied:

1. Face Cropping & Alignment

- Normalize face position and orientation
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2. Background Removal

- Reduce noise using tools like rembg
-

3. Sketch Generation

- Convert images → sketches using:
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- Edge detection (Canny)
-

- Pencil sketch filters
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4. Resizing

- Standard size: **256 × 256 pixels**
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5. Normalization

- Pixel scaling to range **[-1, 1]**
-

6. Pair Creation

- (Sketch, Real Image) pairs for supervised learning
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Augmentation Strategy

To improve generalization:

- Horizontal flipping
 - Random cropping
 - Rotation ($\pm 10^\circ$)
 - Brightness and contrast adjustment
 - Noise injection (to simulate rough sketches)
-

👉 Special augmentation:

- **Sketch-style variation augmentation** to mimic different drawing styles
-

Train / Validation / Test Split

A standard split will be used:

- **Training:** 70%
 - **Validation:** 15%
 - **Testing:** 15%
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Rationale:

- Ensures sufficient training data for GAN stability
 - Validation prevents overfitting
 - Test set ensures unbiased evaluation
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2. Secondary Dataset (Recent & Relevant)

CUFS Dataset

Widely used benchmark for sketch-to-face tasks.

Key Details:

- ~606 identities
 - Paired sketch-photo images
 - High-quality alignment
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Why Used:

- Directly relevant to forensic sketch translation
 - Helps model learn **real sketch-to-face mapping**
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Limitation:

- Small dataset size
 - Limited diversity
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CelebA-HQ Dataset

An enhanced version of CelebA.

Key Details:

- 30,000 high-resolution images (1024×1024)
 - Better visual quality
-

Why Used:

- Improves **image realism and texture quality**
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Limitation:

- No sketch pairs (needs synthetic generation)
-

3. Backup Dataset (Mandatory Safety Option)

FFHQ Dataset

Key Details:

- 70,000 high-quality images
 - Diverse demographics (age, ethnicity, lighting)
 - Resolution: 1024 × 1024
-

Why This Backup:

- More **real-world diversity** than CelebA
 - Excellent for **generalization testing**
-

Usage Plan:

- Generate synthetic sketches
 - Fine-tune model if CelebA underperforms
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Section 7: Proposed Methodology

7.1 High-Level Pipeline

The pipeline flows as follows: Raw Input (photo or sketch) → Preprocessing & Edge Extraction → [MLP Baseline / CNN Enhancer / LSTM Refiner] → Pix2Pix cGAN Generator (U-Net) → Generated Face Image → VGG-16 Perceptual Loss Head (Transformer attention) → Discriminator (PatchGAN) → Backpropagation → Deployed Streamlit App.

A hand-drawn diagram of this pipeline is attached as Appendix A. The diagram shows data flow arrows, module boundaries, and loss signal paths.

7.2 Architecture Integration

7.2.1 MLP / ANN — Baseline Model

Role: Establishes a performance floor against which GAN results are compared. A 5-layer MLP (input: flattened 256×256 edge map = 65,536 nodes → 2048 → 1024 → 512 → 256 → output: 196,608 nodes for RGB face) is trained with MSE loss. Expected FID: >200. This demonstrates why structural architectures are necessary.

7.2.2 CNN — Edge Map Enhancer

Role: Before feeding a hand-drawn sketch to the generator, a lightweight 6-layer CNN (3 conv + 3 deconv) cleans noise, closes broken lines, and normalises stroke width to match the training distribution. Trained on CUFS pairs with L1 loss.

7.2.3 LSTM — Sequential Sketch Refinement

Role: Models sketch construction as a sequence of refinement steps. The edge map is divided into 16×16 patches processed in raster-scan order by an LSTM (hidden size 512). Output patch embeddings are reassembled and passed to the generator. This module is optional at inference (can be bypassed for single-step input) but improves results on freehand multi-stroke sketches.

7.2.4 Transformer / Attention — Perceptual Loss Head

Role: Instead of a fixed VGG perceptual loss, a 4-head self-attention Transformer (4 layers, d_model=512) computes semantic similarity between generated and real face feature maps extracted at VGG-16 layer conv4_3. This provides a differentiable, attention-weighted perceptual loss that emphasises structurally important regions (eyes, nose outline).

7.2.5 GAN (Pix2Pix) — Core Generator

Role: The primary generation model. Generator: U-Net with 8 encoder blocks (Conv-BN-LeakyReLU) and 8 decoder blocks (ConvTranspose-BN-ReLU-Dropout) with skip connections. Discriminator: PatchGAN 70×70 receptive field. Total loss: $L_{\text{total}} = L_{\text{adv}} + \lambda \cdot L_{\text{L1}} + \mu \cdot L_{\text{perceptual}}$, where $\lambda=100$, $\mu=10$ (ablation will tune these).

7.3 Transfer Learning

Pretrained model: VGG-16 trained on ImageNet (torchvision.models.vgg16). Frozen layers: all convolutional layers up to and including conv4_3 (features[0:24]). Fine-tuned layers: conv5_1 through conv5_3 (features[24:31]) at learning rate 1e-5. Rationale: Early VGG layers capture low-level textures (edges, colour blobs) that are domain-agnostic; fine-tuning later layers allows the network to adapt its high-level face feature representations.

7.4 Loss Functions

- Adversarial loss (BCE): Trains generator to fool discriminator; trains discriminator to distinguish real from fake.
- L1 pixel loss: Penalises pixel-level deviation from ground truth; prevents blurriness and mode collapse.
- Transformer perceptual loss: Penalises semantic feature-map deviation at VGG conv4_3; preserves structural identity cues.

7.5 Optimizer Comparison Plan

Both optimizers will be run with identical hyperparameters (lr=2e-4, batch=8, 50 epochs) and their training loss curves, validation FID, and final test FID will be compared in a table in the final report.

Optimizer	Learning Rate	β_1 / Momentum	Expected Strength
Adam	2e-4	$\beta_1=0.5$, $\beta_2=0.999$	Fast convergence, handles sparse gradients well
RMSProp	2e-4	momentum=0.0	Stable for non-stationary objectives; used in original DCGAN

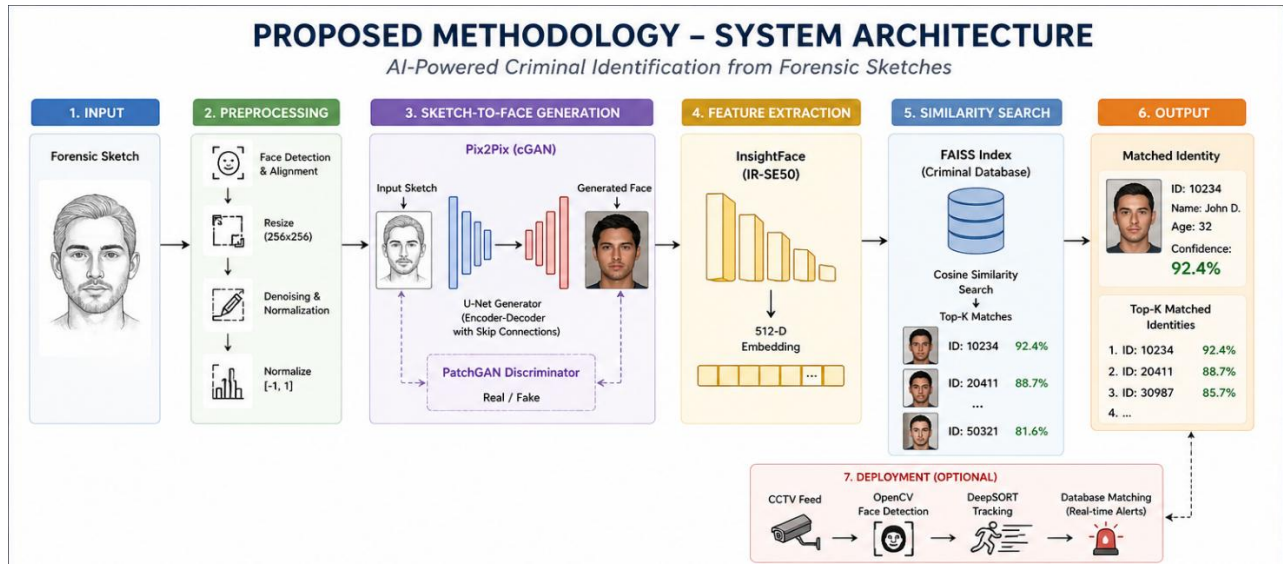
7.6 Regularization Plan

- Dropout (p=0.5): Applied in the first 3 decoder blocks of the U-Net generator.
- BatchNorm: Applied after every convolutional layer except the generator output and discriminator input.
- L2 weight decay (1e-4): Applied to discriminator optimizer to prevent memorisation.
- Early Stopping: Halt training if validation FID does not improve for 5 consecutive epochs.

7.7 Hyperparameter Tuning Strategy

Optuna (v3.x) with TPE sampler and 30 trials over the following search space: learning rate [1e-5, 1e-3] (log-uniform), λ_{L1} [50, 200], $\mu_{\text{perceptual}}$ [1, 20], batch size {4, 8, 16}. Pruning via MedianPruner after epoch 10. Best trial hyperparameters will be used for all final reported experiments.

Section 8: System Architecture



Section 9: Technology Stack

Category	Library / Tool	Version	Justification
Core DL	PyTorch	2.1.0	Primary framework; dynamic graph, strong GAN support
Core DL	TorchVision	0.16.0	VGG-16 pretrained weights, image transforms
Image Processing	OpenCV (cv2)	4.8.0	Canny edge detection, GaussianBlur, resizing
Image Processing	Pillow (PIL)	10.0.0	Image I/O, format conversion
Numerical	NumPy	1.24.0	Array ops, normalization, seed control
Visualisation	Matplotlib	3.7.0	Training curves, output grids
Hyperparameter Tuning	Optuna	3.3.0	TPE sampler, MedianPruner, study persistence
Evaluation	pytorch-fid	0.3.0	FID computation using Inception-v3 features
Deployment — UI	Streamlit	1.25.0	Rapid web UI for sketch upload and face display
Deployment — API	FastAPI	0.100.0	Async REST endpoint for model inference
Deployment — Server	Uvicorn	0.23.0	ASGI server for FastAPI
Environment	Conda	23.x	Dependency isolation; environment.yml committed to repo
Version Control	Git + GitHub	—	Source control; repo: github.com/MoazzamSharif/sketch-to-face-dl
Compute — Training	Google Colab Pro	T4/A100 GPU	Free/low-cost GPU for 50-epoch training runs
Compute — Inference	CPU (local)	—	Inference latency <10s on CPU; no GPU needed at runtime
Checkpoint Storage	Google Drive	—	Colab-native; 15 GB free storage for .pt files
Notebook	Jupyter Notebook	6.5.x	Interactive development and experiment documentation

Section 10: Six-Week Timeline

Week	Focus	Deliverables	Owner	Dependencies
Week 1	Roadmap & SRS	This document; GitHub repo created; folder structure committed; environment.yml drafted	All	None
Week 2	Data & Baseline	CelebA downloaded; Canny preprocessing pipeline complete; MLP baseline trained and evaluated; EDA notebook committed	Bilal (data), Muaz (MLP)	Week 1 complete
Week 3	Core Architecture	U-Net generator + PatchGAN discriminator implemented; first training run on 10K subset; LSTM sketch-refiner prototype	Moazzam (GAN), Muaz (LSTM)	Week 2 complete
Week 4	Advanced Components	VGG-16 transfer learning integrated; Transformer perceptual loss implemented; Optuna hyperparameter search run; CNN edge enhancer trained	Moazzam (perceptual), Bilal (tuning)	Week 3 complete
Week 5	Evaluation & Analysis	Full training on CelebA 162K; FID computed on test set; ablation study (with/without perceptual loss); error analysis; confusion visualisations	All	Week 4 complete
Week 6	Deployment & Report	Streamlit + FastAPI app deployed; demo video recorded; final report written; presentation slides prepared; repo cleaned and tagged v1.0	Muaz (deploy), Moazzam (report)	Week 5 complete

Buffer: Each week includes one unscheduled day (Friday) reserved for overflow, debugging, and unexpected blockers. Critical path: Weeks 3 → 4 → 5 form the training pipeline; any delay in Week 3 directly impacts Week 5 evaluation.

Section 11: Evaluation Metrics

Metric	Type	Target	Justification
Fréchet Inception Distance (FID)	Quantitative (lower is better)	≤ 50	Standard GAN quality metric; measures distance between real and generated feature distributions
Structural Similarity Index (SSIM)	Quantitative (higher is better)	≥ 0.65	Measures luminance, contrast, and structural similarity; more perceptually meaningful than PSNR
LPIPS (Learned Perceptual Image Patch Similarity)	Quantitative (lower is better)	≤ 0.30	AlexNet-based perceptual distance; correlates better with human judgment than pixel-level metrics
Generator Loss (L_{total})	Training diagnostic	< 0.5	Monitors convergence; not a final quality metric
Discriminator Accuracy	Training diagnostic	40–60%	Healthy GAN: discriminator should not be too confident in either direction
Human Perceptual Study (AMT-style)	Qualitative	$> 25\%$ fooling rate	20 annotators rate realism of 50 generated faces on 1–5 scale; matches Isola et al. evaluation protocol
Confusion Matrix on Gender Attribute	Qualitative	—	VGG face attribute classifier applied to generated faces; checks whether demographic distributions match training data
Attention Visualisation	Qualitative	—	Visualises Transformer attention weights over facial regions to confirm model focuses on semantically relevant areas

Section 12: Expected Outcomes & Deliverables

Artifact	Format	Location	Due
Trained Pix2Pix Generator Weights	pix2pix_generator.pt	Google Drive + GitHub Releases	End of Week 5
Full Source Code Repository	Python + Jupyter Notebooks	github.com/MoazzamSharif/sketch-to-face-dl	Week 6
Streamlit + FastAPI Deployed Demo	Public URL (Streamlit Cloud)	TBD at deployment	End of Week 6
environment.yml (pinned deps)	YAML	GitHub /repo root	End of Week 2
Phase 1 SRS Document (this doc)	PDF / DOCX	LMS + GitHub /docs	April 27, 2026
Final Report (Phase 6)	PDF	LMS	End of Week 6
Presentation Slides	PPTX	LMS + GitHub /docs	End of Week 6
Demo Video (3–5 min)	MP4	YouTube (unlisted) + GitHub	End of Week 6
Experiment Logs	CSV + Optuna SQLite DB	GitHub /experiments	Ongoing

Section 13: Risk Analysis & Mitigation

Risk	Probability	Impact	Mitigation Plan
Insufficient compute — Colab session timeouts interrupt long training runs	High	High	Use Colab Pro; save checkpoints every 5 epochs; implement resume-from-checkpoint logic in training loop
CelebA dataset unavailable or download broken	Medium	High	FFHQ backup dataset identified (Section 6.3); preprocessing pipeline is identical; switch takes < 1 hour
Generator mode collapse — output diversity collapses to single face	Medium	High	Monitor discriminator accuracy; add instance noise to discriminator inputs; use spectral normalisation; reduce λ_{L1} if collapse detected
Model not converging — loss oscillates without improvement	Medium	High	Implement learning rate scheduler (CosineAnnealingLR); try RMSProp if Adam fails; reduce batch size; re-initialise with orthogonal weight init

Team member unavailability (illness, exam clash)	Low	Medium	Every task has a secondary owner (Section 14); critical-path tasks (Weeks 3–5) are assigned to Moazzam as primary with Bilal as backup
Scope creep — adding identity preservation or high-resolution output	Medium	Medium	Scope explicitly limited to 256×256, non-identity-preserving generation (Section 5.2); any new feature requires group consensus and instructor approval
Integration failure during deployment — model runs in Colab but crashes in FastAPI	Low	High	Test FastAPI endpoint with exported ONNX model in Week 5 before full deployment in Week 6; maintain a fallback Gradio app as deployment alternative
Dataset ethics — CelebA faces may include unconsented public figures	Low	Medium	CelebA is cleared for non-commercial academic research; all generated faces are synthetic and not linked to real identities; no real faces published in demo

Section 14: Team Roles & Responsibilities

Area	Primary Owner	Secondary Reviewer	Deliverables Owned
Project Lead & Architecture	Moazzam Sharif (S2024CS009)	Bilal Asif	U-Net generator, PatchGAN discriminator, Transformer perceptual loss, final report
Data Engineering & Preprocessing	Bilal Asif (S2024CS005)	Muaz Ahmed	CelebA download/preprocessing pipeline, CUFS alignment, augmentation code, EDA notebook
Secondary Architectures (LSTM, CNN Enhancer)	Muaz Ahmed (S2024CS016)	Moazzam Sharif	LSTM sketch-refiner, CNN edge enhancer, MLP baseline
Hyperparameter Tuning & Evaluation	Bilal Asif (S2024CS005)	Moazzam Sharif	Optuna study, FID/SSIM/LPIPS computation, ablation table
Deployment (Streamlit + FastAPI)	Muaz Ahmed (S2024CS016)	Bilal Asif	FastAPI endpoint, Streamlit UI, demo video, deployment documentation
Documentation & Version Control	Moazzam Sharif (S2024CS009)	Muaz Ahmed	SRS document, final report, README, GitHub repo structure, presentation slides

Section 15: References

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[4] Z. Liu, P. Luo, X. Wang, and X. Tang, "Deep Learning Face Attributes in the Wild," in Proc. IEEE ICCV, pp. 3730–3738, 2015.

[5] H. Wang, Y. Hu, and J. Li, "Face Sketch Synthesis via Semantic-Driven Generative Adversarial Network," IEEE Trans. Image Process., vol. 32, pp. 1415–1428, 2023.

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[10] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A Next-generation Hyperparameter Optimization Framework," in Proc. ACM KDD, pp. 2623–2631, 2019.

Section 16: Appendices

Appendix A — Pipeline Diagram (Placeholder)

[Insert hand-drawn pipeline diagram: Input Sketch → Preprocessing → CNN Enhancer / LSTM Refiner → U-Net Generator → PatchGAN Discriminator → Generated Face → Perceptual Loss → Backprop]

Appendix B — System Architecture Diagram (Placeholder)

[Insert system architecture diagram: Browser (Streamlit) → HTTP POST → FastAPI Endpoint → PyTorch Model Server → JPEG Response → Browser Display]

Appendix C — Proposed Repository Folder Structure

```
sketch-to-face-dl/
├── data/                # Raw and processed dataset (gitignored)
├── notebooks/           # Jupyter notebooks (EDA, training, evaluation)
├── src/
│   ├── model.py         # Generator, Discriminator, LSTM, CNN Enhancer
│   ├── train.py         # Training loop, optimizer comparison
│   ├── evaluate.py      # FID, SSIM, LPIPS computation
│   └── preprocess.py    # Canny pipeline, augmentation
├── app/
│   ├── main.py          # FastAPI endpoint
│   └── streamlit_app.py
├── checkpoints/         # .pt model files (gitignored, stored on Drive)
├── experiments/         # Optuna DB, training logs
├── docs/                # SRS, final report, slides
├── environment.yml
└── README.md
```